

**ENHANCING DATA-DRIVEN INTERVENTION AND ACADEMIC
TRACKING IN A HIGHER EDUCATION INSTITUTION
THROUGH THE DIGITAL SYSTEM**

A Project Proposal

Presented to

The Faculty of the Graduate School

University of the Immaculate Conception Davao City

In Partial Fulfillment

of the Requirements for the Degree

Master of Arts in Education

major in Information Technology Integration

Zolina, Melissa T.

January 2025

CHAPTER 1

INTRODUCTION

Background of Study

Digital system is a computerized academic monitoring system, which is aimed at tracking the student attendance, classroom activities score and providing analysis that allow detecting students at risk (Sharma et al., 2023). Prior to automation, attendance recording is commonly done through manual methods, which are prone to errors, proxy attendance, loss or misplacement of attendance sheets, and difficulties in tracking students attendance performance over time (Arif, 2018). Automated attendance and academic monitoring systems can greatly enhance accuracy, decrease the volume of work done by teachers and give real-time information (Raj et al., 2024). It is also demonstrated that QR code-based and automated attendance systems help reduce proxy attendance and simplify the process of class management because they are faster and more reliable to use (Okokpujie et al., 2018; RSIS International, 2023). Moreover, the studies of analytics-based academic systems point out the fact that dashboards equip schools with improved early warning of struggling learners by combining attendance, performance, and engagement data into actionable visual reports (IARJSET, 2025; Irjets, 2024). The attendance system described by Arif (2018) requires extended time to operate while students can use proxy signing to mark absent classmates as present. The implementation of manual attendance systems requires teachers to spend 10 to 15 minutes from teaching time for each class according to David (2025).

Globally, International schools all over the world are taking the initiative of going digital to modernize academic tracking. KIS International School in Bangkok implements ManageBac+ to enable the teachers to take care of attendance and monitor classroom activities effectively (KIS International School, n.d.). Equally, International School Bangkok (ISB) uses a student information platform known as PowerSchool, which allows real time digital tracking of student attendance and grade level classroom related information (International School Bangkok, n.d.). Similarly, UWC South East Asia (UWCSEA) also uses institutional portals like CIMS to handle student records and attendance record (United World Colleges, n.d.).

In the Philippines, various institutions of higher learning have started to implement digital attendance and academic monitoring systems, but this has not been uniformly applied to institutions. An interesting case is the Lyceum of the Philippines University-Laguna (LPU-Laguna), a higher education institution based in Calamba City, Laguna, having created the LPU-Laguna SEALS electronic attendance and logging system that uses technology with notifications, and web-based portal to give real-time access to student attendance (LPU-Laguna, 2016). Other local studies show the same improvements, including attendance monitoring with alerts that help to enhance accuracy and parental involvement (IIARI, 2022), and analytics-based monitoring systems in educational institutions to promote tracking of learners and facilitate early interventions (De La Salle University, 2020; EJournal, 2019). Notwithstanding these technological efforts, the existing literature suggests that most of the private schools in the country still rely on the manual or semi-digital processes, which results in delays, data inconsistency, and a low level of capability to identify the at-risk students in good time (WJARR, 2023).

To overcome these difficulties, this research proposes the Digital System to record and monitor attendance, classroom activities, and academic records in private colleges. While existing studies have explored automated attendance tools, analytics dashboards, and early-warning mechanisms, there remains a gap in integrating these components into a single, user-friendly academic tracking system tailored to the needs of higher education institutions in rural or semi-urban Philippine contexts. Moreover, limited research has examined how such a unified digital platform can simultaneously support instructors' data-driven interventions and improve student monitoring at the local institutional level. The researcher will present the study to the institution's administrators and faculty members to inform improvements in academic monitoring and early intervention practices.

This study is aligned with UN quality education 4 Sustainable Development Goal by focusing on inclusive and equitable quality education and lifelong learning opportunities for all, UN sustainable development Goal number 9 is also associated for innovation. This study deals with the development and acceptability of a digital system for recording student attendance, classroom activity scores, and generating analytics to identify at-risk students in a private college institution in SEAIT Tupi, South Cotabato.

The results are to be presented in academic forums and in-house presentations for school to adopt and as a paradigm of how to bring cheap and effective application of technology to the academic world. In this way, I contribute to both academic and practical knowledge of education and IT integration. The findings will also be shared through research forums, conferences, and academic presentations within and outside the region. In addition, the researcher will aim to

publish the results in peer-reviewed journals to contribute to broader scholarly discussions on educational technology integration.

Statement of the Problem

This study aims to develop and evaluate a Digital System for academic tracking in a private college institution in Tupi, South Cotabato. To address the inefficiency of the current manual academic tracking methods and a lack of real time tools for student intervention. Specifically, this study will answer the following questions:

1. What are the technical requirements and architectural design of the Digital System in terms of:
 - a. Data Input and Integrity;
 - b. Analytics Dashboard;
 - c. Security and Privacy features?
2. What is the level of software quality of the developed Digital System as evaluated by IT experts based on ISO 25010 standards.
3. How does the Digital System facilitate the data-driven intervention practices of instructors in terms of:
 - a. Identification of at-risk students (Attendance & Grades);
 - b. Timeliness of intervention; and
 - c. Accessibility of student records?
4. What is the level of acceptance of the Digital System among instructors and students based on the Technology Acceptance Model (TAM) in terms of:
 - a. Perceived Usefulness;
 - b. Perceived Ease of Use; and
 - c. Intention to Use?

5. What types of intellectual property (IP) protection can be drafted and filed for the developed application?

Review of Related Literature

In this section, the literature regarding the Digital System is examined with the emphasis on digital data on the attendance of students, classroom activities, and risk identification of students through analytics. It talks about the role of these systems in facilitating academic tracking, enhancing teaching effectiveness and promoting decision-making grounded on data.

Academic Recording

The effective management of academic records is critical for smooth educational system operations and academic efficiency. As part of the Guidance Office at Jose Rizal Memorial State University, Acosta et al. (2021) discovered an Automated Student Record System to minimize the duration to process, decrease human error, and ease access to the files of students. Automatic systems are less time-consuming in the management of records and are more reliable in providing accurate data as compared to manual systems.

Automated Student Academic Record Management System designed by Salangsang and Divina (2020) based on the Business Intelligence approach that revealed the shift in decision-making based on the data integration, with an equivalent reporting and analysis tools, better reports with the information on long-term planning. The results showed that digital academic record-keeping would facilitate documentation and upgrade the institutions' capacity to access and use students' data for the purpose of improving learning.

These investigations have revealed that the automated recording of academic performance can, in effect, raise the accuracy, openness, and efficiency, which in turn would enable the institutions to better inform their policymaking and academic intervention.

Data Input and Integrity. Effective academic information systems involve data entry and integrity because there should be no erroneous or conflicting data that can directly affect the institutional decision-making, the accuracy of reporting, and monitoring of students. Global studies have continuously indicated that manual data entry (i.e. roll calls, paper logs or incomplete spreadsheets) is still very vulnerable to human error, irregularities, and information loss. According to Collabco (2023) Student Attendance Monitoring Guide, manual input has been identified as a primary source of incomplete, duplicated, and inaccurate student records and automated systems are necessary to provide reliability and consistency. When discussing the digital data management in the scope of the bigger picture, Konig et al. (2024) emphasize that paper-based records are more likely to be characterized by transcription and corruption, while electronic entry systems enhance traceability and minimize the cases of data corruption. On the same note, Hall et al. (2020) documents that automated data-capture systems lead to a high level of data integrity within institutional settings through their standardization of input formats and performing data validation during collection.

New literature on technology also highlights the fact that record accuracy should be maintained through real-time and automated data input. According to VerifyEd (2023), QR-based attendance systems minimize the presence of false or proxy entries since each scan is associated with a singular digital identity and this can increase the integrity of attendance records. Digital attendance studies also

substantiate that automated input eradicates all the frequent issues that are contained within the manual systems, including illegibility to read, absent entries, and unauthorized modifications (Kumar & Verma, 2024). The MIHCM (2023) industry analyses also prove that centralized cloud-based academic records with automated data input offer greater protection against data loss, tampering, and duplication, which enhance greater integrity and institutional reliability.

Research in related areas provides further support to the need to shift data entry, which is manual, to the digital realm. According to the Data Curation Centre (2023), the inherent qualities of electronic record-keeping systems help to achieve a higher level of data integrity due to time-stamped logs, controlled access of users, and audit trails - functions that are not possible with the paper-based systems. Similarly, the paper by Sahoo and Raj (2019) on electronic academic systems reveals that digital entries reduce the number of human-related inconsistencies considerably and provide more stable, verifiable and secure records to support the institutional audit and performance review of students.

The significance of automated data input in the academic environment is also confirmed in local and regional studies. In a Philippine study, Mendoza et al. (2024) found that institutions that use manual encoding experience delays, inconsistency in reports, and errors build up, particularly in the process of attendance and grading. Their results indicate that automated academic platforms with digital input also help in increasing accuracy and decreasing administrative load.

Digital systems demonstrate that the guarantee of the correctness of data input and the integrity of data implies the shift to the automated and digitally validated systems. Automated systems reduce the risks of errors, manipulation,

and consistency and improve the academic tracking transparency. The findings are a good predictor of the formulation of the Digital System that attempts to introduce automated data entry and reliability systems in the attendance tracking process, the classroom activity logs and control over the academic records of a private college institution.

Analytics Dashboard. Dashboard analytics now form an imperative part of the modern educational systems and serve to aid real-time monitoring, intervening early and also make decisions based on data. The use of dashboards in terms of international studies has shown that it greatly improves the capacity of the institution in terms of identifying trends in the behavior and performance of students. According to Solved Consulting (2023), AI-driven analytics may be utilized to detect at-risk students by examining their patterns of attendance, participation, and submission, which allows students to be supported in time. Similarly, XenonStack (2023) emphasizes the fact that predictive analytics dashboards enhance student retention by combining multi-source data, including attendance, grades, and behavioral indicators, to predict the risk of dropout and implement specific intervention. A study conducted by the Attendance Works (2020) also indicates further that attendance dashboards can help improve the monitoring of chronic absenteeism, especially in remote and blended school settings, as it allows visualizing the indicators of engagement and responding proactively to the problems related to attendance in schools.

The recent academic sources also focus on the practicality of dashboards in enhancing student interaction and education. Empirical research conducted by de Oliveira et al. (2024) concluded that interactive learning analytics dashboards go a long way in increasing student motivation and self-regulation by allowing them

to readily access visual summaries of performance and engagement rates. Likewise, Palos Publishing (2023) illustrates that dashboards with heatmaps of engagement and visualizations of activity allow instructors to recognize disengaged learners and change the approach to the instruction. It is also reported by Forecast Global (2022) that student-performance dashboards that were deployed in higher education institutions enhanced the efficiency of instructors in tracking gains, detecting learning lags, and launching early teaching interventions. All of this makes dashboard analytics one of the key instruments towards empowering academic monitoring and improving the quality of education.

On institutional levels, dashboard systems have become part of academic management systems. According to AcademyOcean (2023), the accuracy of academic decision-making is enhanced with analytics dashboards, which allow integrating attendance logs, activity records, and assessment results into one interface that can be easily understood by an administrator and an instructor. SchoolTool Analytics (Mindex, 2023) underlines the importance of the fact that dashboards with visualization tools, i.e. trend lines, comparative graphs, and cohort tracking, allow schools to evaluate the patterns of performance and distribute resources efficiently. This is furthered by Practicum Analytics (2023) which demonstrates that dashboards aid in adaptive learning through constant tracking of student performance, thus facilitating one-on-one learning instruction based on real-time data.

Combined, the literature reviewed makes dashboard analytics an innovative element of academic information systems, as it provides real-time tracking, predictive analytics, and improved support in teaching. The features are directly applicable to the creation of the Digital System that will introduce the dashboard

features to present the attendance patterns, logs of classroom activities, and indicators of academic performance. The implementation of analytics dashboards will help educators see the trends, identify academic risks in a timely manner, and facilitate more informed and better-informed and data-driven choices that will help to increase the student outcomes in the context of the private higher education.

Security and Privacy features. Global literature underlines that academic performance, attendance, and at-risk analytics digital systems should focus on excellent security and privacy to ensure that the data of students remains confidential. Attendance systems based on biometrics like fingerprint verification have been reported to massively enhance data integrity and block unauthorized access with the reported institutions noting a quantifiable decrease in fraudulent records entries (Kumar & Khan, 2024). International standards on the management of student records also emphasize the importance of encrypted storage, limited access, and minimization of the data to avoid violations and abuse (CPD Online, 2024; IAPP, 2022).

Learning analytics system research shows that early-warning dashboard systems must have strong privacy safeguards, such as pseudonymization and role-based access control (RBAC) to avoid excessive exposure of risk data of students. The ACM-published research revealed that systems without RBAC had 32 times higher risks of exposing unauthorised data during the analytics access (Chen et al., 2019). Architectures that are cloud-native and microservice also contribute to the enhanced security of the system through perimeter isolation of the application components, making the use of container security frameworks lower attack surfaces on deployments (Cloud Security Alliance, 2024). The fact that in the international context, educational early-warning systems are

implemented with the help of encrypted communication channels and adherence to GDPR indicates that the Internet-based systems provide a significant boost in the trust that stakeholders have and the acceptability of the systems (PowerSchool, 2023; GeeksforGeeks, 2024).

Increasing use of secure student information systems that integrate encrypted data storage, secure APIs, and structured access controls are emphasized as a way of protecting academic data presented in National literature. An attendance and student monitoring system in the Philippines showed that the use of hashed identifiers and access privileges based on tier of access enhanced the integrity of academic and attendance records (ISJEM, 2023). Moreover, as per a recent audit of records management in schools in Southeast Asia, the digital analytics dashboard should restrict the exposure of sensitive indicators (such as risk scores) to staff that are authorized to access this information only in order to meet institutional privacy demands (Dan Frederking, 2023).

A study concerning early-warning systems in the Philippines also reveals that systems with built-in secure analytics pipelines can assist the institutions to act at an earlier stage, yet still comply with the expectations of student privacy. Safe notification systems, including encrypted messages that are only delivered to authorized teachers and students, were found as the key features of responsible student monitoring systems (ERStrategies, 2023). These papers point out to the increased necessity of privacy-by-design designs in educational institutions.

Local research is consistent with international reports, which point out that a secure digital system of attendance and academic tracking should have data protection measures in order to avoid unauthorized usage of the system. End-to-end encryption between teacher dashboards and institutional servers, which was

reported by a locally developed student data management platform, minimized the risk of data interception by up to 65 percent (Kiteworks, 2024). Further, the school-level system design research highlights the ineffectiveness of out-of-date record-keeping principles and recommends the use of encrypted relational databases, role-based access, and adherence to the FERPA-equivalent standards of student protection (Back4App, 2024; Scribd SIS Design, 2019).

Also, local deployments of cloud-based academic monitoring systems discovered that microservices architecture improves security and scalability and reliability through the isolation of sensitive resources, including attendance logs and academic score databases (RSIS, 2024). Institutional privacy policies in the area also emphasize that acceptable data retention, audit trails, encrypted backups, and the least-privilege principle are necessary to prevent unintentional student data leaks (CypherLearning, 2024; Wisconsin EDU, 2023). In both international, national and local literature, it is unanimously agreed that security and privacy are non-negotiable elements of digital academic systems, particularly those ones that gather attendance records, performance scores and risk analytics. These are often mandated by such features as encrypted data storage, secure authentication (biometrics or multi-factor), RBAC, secure APIs, anonymized analytics, and adherence to regulations, including GDPR, FERPA, COPPA, and institutional data policies. The insights directly shape the Digital system as developed as an architectural design to guarantee that the platform safeguards sensitive data about students and provides the capacity to accurately track attendance, monitor activities, and offer at-risk learners early-warnings analytics.

Software Quality Based on ISO 25010 Standards

The ISO/IEC 25010 software quality model offers an extensive model to assess the quality of software products in regard to eight attributes, functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. This model is popular in contemporary software evaluation since it has a structured and globally accepted standards of establishing whether a system is fulfilling the demands of users and the organization (ISO 25010, n.d.). According to Codacy (2023), ISO 25010 allows the developers and evaluators to evaluate the software both functionally and non-functionally, to make sure that the systems are efficient in performing the tasks they need to do, but also ensure that they remain user-satisfied, secure, and functional. In a similar vein, Perforce (2024) mentions that ISO 25010 can be utilised as a standard of assuring the robustness of software and even long-term reliability especially in systems designed to be used by institutions.

In accordance with recent research, it has been proven that ISO 25010 can be employed to assess different software systems, including those related to education and administration. Hanggoro et al. (2024), carried out a systematic literature review of the low-code and no-code development platforms using ISO 25010 evaluated the software quality of these environments and discovered that the model is suitable in testing new technologies. The article by Puspaningrum et al. (2017) used the ISO 25010 to determine the functional suitability of an Academic Information System (AIS) and the results stated that the model generated the precise and goal-focused measures in accordance with the needs of the institution. Their results prove the appropriateness of ISO 25010 to academic systems where the accuracy, responsiveness, and reliability are essential.

The application of ISO 25010 in educational software is also supported in other empirical studies. Setyanto et al. (2022) evaluated an Android-based learning media application according to ISO 25010 and found that the system fulfilled the significant quality features, in particular, the usability and portability, which supports the model in its flexibility to evaluate the mobile learning technologies. Peters and Aggrey (2020) used ISO 25010 to assess the ERP systems in higher institutions of learning, which proved that the framework can be scaled to large-scale institutional platforms. Their results demonstrate the strength of the framework to evaluate the complex systems that process an enormous amount of academic and administrative data.

Its effectiveness in the assessment of systems offered to monitor, track academic, or manage administrative systems is also supported by local and regional studies based on the ISO 25010. Laudza and Sofyan (2024) tested an academic information system by using the ISO 25010 and concluded that the majority of the quality characteristics were at an acceptable level, but security and maintainability had to be improved- delivering an idea of how the model can be used to find areas where the system is weak. According to a recent study by De Vera et al. (2023), a QR code-based smart attendance monitoring system was evaluated by using ISO 25010; the study has indicated that the system performed well in functional suitability, performance, and usability. Their study shows that the model is applicable to the systems dedicated to the attendance tracking and the digital monitoring-aspects directly related to the goals of the Digital System.

In general, there is evidence in the literature that ISO/IEC 25010 has been a strong and interoperable scheme to evaluate software quality, specifically within the educational and monitoring systems. Its wide validation in various studies

including the study of academic information systems and mobile learning platforms and apps that monitor attendance, among other things, justify its suitability in the assessment of the Digital System. Application of ISO 25010 enables the Digital System to be evaluated through the recognized standards globally thus making the design, functionality, and performances of the Digital System to meet the expectation of the institution regarding accuracy, usability, reliability, and security.

Digital System Identification of At-Risk Students

The combination of attendance/academic performance data into early-warning systems has been well reported as great tools to help reveal at-risk of dropout or failure students and provide timely interventions. Evidence on large scale (randomized trials) indicates that scalable early-warning system can be applied in thousands of schools to make dropouts less likely and enhance retention; a study on an early-warning intervention conducted by the World Bank involved 4,000 schools and found that monitoring and targeted support could be measured (World Bank, 2021). The synthesized research indicates that combining attendance and grade-data in dashboards allows schools to triage students into levels of support and allocate resources in a more effective way (ErStrategies, 2023). AttendanceWorks and associated syntheses highlight that attendance is among the best initial predictors of subsequent academic underperformance, and that dashboards that showcase chronic absence combined with grades enable practitioners to detect at-risk pupils earlier than grade indicators do (AttendanceWorks, n.d.; Every1Graduates, 2019).

The technical research indicates that simple models with attendance and grade markers are strong predictors with risk identification in the short term. As an example, machine-learning and logistic-regression can be used combining

previous attendance history and course grades in order to identify the students who will need intervention in the following term, with reasonable accuracy (EdWorkingPapers, 2024; ACM studies on predictive analytics, 2024). Evidence in the field is found in practitioner platforms (Panorama, PowerSchool, SchoolAnalytix) where it is stated that schools with combined attendance-and-grade dashboards report quicker identification of students requiring Tier-2 supports and better responsiveness of counselors and teachers (PowerSchool, 2023, Panorama Education, 2021, SchoolAnalytix, n.d.).

Local guides and manuals propose the integration of attendance measures with academic performance to be the most successful early-warning strategies. The protocols of tiered intervention described in the Every1Graduates early-warning manual (Every1Graduates, 2019) and in other practitioner reports include automated dashboard flags (attendance declines, failing grades) and follow-up by teachers and specific supports. Medical and educational research studies record the fact that attendance can frequently be an early indicator of grade drops, and thus systems that continually track both indicators will give earlier and more proactive warnings (PMC systematic reviews, 2023; 2024). Case studies of schools that have signed contracts with school analytics vendors have shown that when teachers are alerted in dashboards of combined attendance-and-grade risk, teachers make more timely contact (emails, phone calls, office hours), and schools record more rates of student re-engagement within weeks of contact (PowerSchool, 2023; SeatsOne.ai, 2024).

A number of practical toolkits are workflow-focused: automated notifications (to teacher and student), teacher-initiated contact within 48-72 hours and brief follow-up cycles to determine whether the student re-engages (PBIS Missouri Tier-

2 guidance; EdStrategies implementations). These procedural recommendations find their reflection in empirical research that interventions caused by early warning messages (instead of after-term grades) increase the probability of avoiding academic drop (World Bank RCT; ERStrategies, 2023).

Institutional and campus-level Project reports and theses show how caseload prioritization via analytics dashboards based on attendance and grades can be used by faculty and student-support teams. Campus analysis shows that the use of such dashboards accelerates the process of outreach and making at-risk groups visible: one institution case study reported more than 30% of the advisors have made contact earlier after installing the dashboards (institutional implementation reports; Murray State thesis, 2022). It is also found in local practitioner writeups (DreamClass, Panorama blogs) that instructors tend to implement more so-called soft interventions (reminder messages, brief check-ins) when a dashboard displays a negative attendance trend along with low scores on quizzes rather than using the end-of-term grade trigger.

Although dashboards allow a timely result, the literature warns that alerts alone cannot create change, but to be effective requires human response capacity (teacher time, counselor availability) and the quality of intervention protocols (cite Journal of Technology in Education studies; ERStrategies). Other issues are false positives and model bias: predictive models using attendance and grades may mirror structural inequities without supervised validation and combination with human review (AI /ethics literature; PMC articles). The problem of data quality like the absence of the attendance record, late grading also lowers the predictive accuracy, which is why the predictive accuracy should be considered a high priority in any Digital implementation (Integrate.io; ERStrategies).

All evidence points to the positive aspect of incorporating attendance and academic performance into analytics dashboards that can alert both teachers and students, as combined metrics allow revealing the risk sooner and allow to make more timely and specific interventions. Guides on practitioner, empirical reviews, and large-scale trials would suggest workflows that include automated alerting with swift teacher response as well as tiered supports-practices that can be operationalized by the Digital System. Another aspect that is mentioned in the literature to ensure success is the quality of data capture, proven predictive thresholds, the capacity of teacher to respond, and bias and data governance attention. These results explain why the design of Digital was focused on real-time attendance + grade analytics, teacher/student notification, and inherent intervention tracking.

Digital System Timeliness of Intervention

One of the main processes through which early-warning systems (EWS) deliver positive school results is timely intervention: dashboards should not merely identify struggling students but are supposed to cause rapid and coordinated actions (teams, outreach, supports) to be effective (Frazelle and Nagel, 2015; ERStrategies, 2023). Large observational studies and meta-analyses demonstrate that the quicker a school becomes responsive following a flag, the higher the likelihood of re-engaging students: systematic guidance, hence, suggests regular data review intervals and near-real-time notification procedures to allow staff to take an action before risk cascades can occur (Frazelle and Nagel, 2015; de Vasconcelos et al., 2023).

Strict assessments measure the benefits of the systems in combination with the timely measures. The results of a randomized controlled impact study with 73

Midwest high schools (N [?] 37,671 students) showed statistically significant decreases in one year in EWIMS schools compared to controls: chronic absenteeism decreased by 4 percentage points (14% to 10%), and percent failing one or more courses decreased by 5 percentage points (26% to 21%) with EWIMS intervention compared to controls- the authors attribute these decreases to related EWS processes that allowed more timely intervention assignment. The paper also provided a statistically significant yet small positive change in the average GPA (2.98 vs. 2.87) in EWIMS schools (Faria et al., 2017). These findings indicate that EWS in combination with follow-up during the early phase may lead to short-term attendance and course achievements.

Experimental and quasi-experimental studies highlight that timing, as well as fidelity, is important. A mixed-methods pilot study of Ninth Grade Success Initiative (in five high schools in Washington State) reported an increase in course grades and behavioral outcomes and not significant change in attendance but qualitative data reported that timely coaching and frequent Success Team meetings had enhanced services targeting and thus where the cycles of rapid review and intervention were applied, academic results improved (Knight et al., 2022). These assessments point toward the fact that dashboards are not enough: the timely and structured human response (meetings, outreach within days, coaching) is the other crucial component that analytics should have.

In their recent study on relational early-warning models (IAFREE), the importance of timeliness is emphasized by Frontiers in their measurement framework: the expansion of indicators to relational factors (student-school, student-family) allows the IAFREE model to be more sensitive to early relational decline, which predicts attendance or grade drops, and thus an earlier flagging,

which leads to an earlier intervention opportunity at scale (de Vasconcelos et al., 2023). Psychometric validation in their Brazilian sample (N = 15,924) demonstrated that the relational model can predict earlier at-risk signals than traditional ABC indicators, giving more lead time to timely supports.

Practitioner guides and impact syntheses emphasize specific timing measures to take response: automated notifications are to trigger teacher appeals within 48-72 hours, and recorded follow-ups and progress reviews within 2-4 week bars would test the effectiveness of the interventions (Every1Graduates, 2019; ERStrategies, 2023). REL and ERStrategies implementation toolkits suggest forming EWS teams which should hold regular (weekly to monthly) meetings to discuss flags and allocate timely Tier-2 supports, which characterizes the routines as essential to converting dashboard signals into results (Frazelle & Nagel, 2015; ERStrategies, 2023).

Mass evaluations of practitioners support the pragmatism of timeliness. Indicatively, the Connecticut RISE and associated district partnerships reported almost a 10 percentage-point increase in 9th-grade on-track rates in districts that integrated EWS dashboards alongside rapid intervention protocols over a multi-year partnership, indicating the accretive nature of consistent, timely data-to-action loops (ERStrategies, 2023). Learning-analytics implementers also attest that schools that implemented automated alerting and instant teacher outreach recorded quicker student re-engagement and uppermost short-term task completion as compared to those that reviewed alerts with monthly gatherings (PowerSchool, TASO reports).

There are also quantitative evidences that point to restrictions in instances where timeliness is poor. An implementation fidelity study conducted by the REL

Midwest EWIMS found implementation fidelity to be low in the first year: although EWIMS schools were more likely to have a dedicated team and use an EWS, they did not have any difference with controls regarding frequency of data review and number/type of interventions offered- and in those cases, the impacts were smaller on some outcomes (Faria et al., 2017). This demonstrates that effects are a result of timely review and follow through and not system adoption alone.

Local and institutional case studies (university course dashboards, district pilots) capture operational timing measures and the size of short-term effects when dashboards are used to take rapid action. An example of this is the rollout of Penn State's "Course Insights" which made instructor dashboards accessible to all campuses and focused on timely notifying instructors, internal reports indicate that instructors are reporting faster improvements to flagged students and pilot reports indicate that by 30 percent or more of students report improvements in timely outreach after rollout of the dashboard (institutional reports and Course Insights rollout communications). In schools where EWS were incorporated into counselor routines and automated messages, the schools reported more timely check-ins and more assignment bouncing (institutional case studies; TASO impact summaries).

Effective thresholds leading to timely action are also noted in practice where there would be some instances of typical operation triggers: 10 percent decline in attendance, two consecutive missed assignments, or a low score on a high-stakes test: these triggers when combined with automated alerts and same-week teacher messages in one or more pilot districts have been associated with improved short-term engagement outcomes (Every1Graduates, ERStrategies, local reports).

The consistency of the literature is that the timeliness of the intervention is the most important factor between analytics and student improvement: dashboards must generate actionable flags within time so that the staff can take action on them, and the school must have the structured and time-bound processes (teams, 48-72 hour outreach, short follow-up cycles) to transform such flags into the supports (Frazelle and Nagel, 2015; ERStrategies, 2023; Faria et al., 2017). This is supported by empirical effect sizes: EWIMS caused a reduction in chronic absentees by 4 pp and course failure by 5 pp in the first year of operation when EWS processes were present (Faria et al., 2017) and district partnerships that sustaining timely EWS practices brought about gains of about 10 pp in 9th-grade on-track rates over a multi-year intervention (ERStrategies, 2023). Conversely, pilot analyses show that gains are diminished or non-existent in instances where there is slowness of the data-to-action process or lack of fidelity to implementation (Knight et al., 2022; Faria et al., 2017).

These findings would mean three design and implementation priorities at Digital system, specifically: (1) real-time/near real-time analytics that fuse attendance and grade signals, (2) automated notification processes that bring flags to the correct instructor/counselor with recommendations on the next steps, and (3) institutional processes (EWS teams, progress monitoring cycles) to make sure the follow-through is done and followed through. The inclusion of these features in Digital is more likely to cause flags to result in quick, productive interventions that are reflected in the reported advances in district and randomized trials.

Digital System Accessibility of Student Records

Availability of student records- both in technical access to the records and in fair usability by all learners- has become a key issue in the design of educational

technology. LMS accessibility report has systematic reviews that indicate a large variance in standards like WCAG and Section 508 compliance; numerous platforms (Moodle, Blackboard, Canvas, Google Classroom) demonstrate a lack of provisions to learners with visual, auditory, and cognitive disabilities, restricting opportunities of accessing records and dashboards equally (Ifenatuora et al., 2023). The existing studies regarding learning-analytics adoption highlight the additional point that it is not only an equity concern but also a practical necessity to introduce data-driven intervention: dashboards and integrations of SIS should be accessible to educators and students of diverse abilities to promptly implement supportive measures (Mutimukwe et al., 2021; Open University LAK review, 2019). Furthermore, the recent research in the formative assessment platforms indicates that the decision to design mobile and low-bandwidth has a significant impact on whether the students can access attendance and performance records anytime/anywhere, and the availability of mobile interfaces contributes to the enhancement of the access to the records of low-connectivity users (Formative/JMIR, 2025).

Access to electronic student records has been recorded to be different and is an impediment to effective use of data, as evidenced by national surveys and practitioner reports. Means et al. (2007) indicate that in the U.S. in 2004-05 it was only about a half (48%) of the teachers in K-12 who had access to an electronic student data system--reflecting historic disparities that may still be present in certain situations and the necessity to design broad access (Means et al., 2007). SIS and analytics provider guidance focuses on integrating (SIS - LMS - analytics) as the primary architectural consideration of record accessibility: with an integrated set of systems, teachers have coherent views (attendance, grades, participation)

in a single place and can more readily respond to alerts; with siloed systems, teachers must log into several systems and are less likely to do anything with the data (Panorama Education; PowerSchool, 2023). In implementation reviews, some typical technical obstacles such as inconsistent data structure, the unavailability of single sign-on (SSO) and bad mobile responsiveness are noted to minimize the real-world availability of records to students and staff (Innovare blog; CEUR proceedings).

The use of recent case studies and system appraisals in institutions and journals indicate viable strategies and gaps that are still in existence at campus level. When attention to SIS-LMS integration, mobile usage, and adherence to WCAG enhanced teacher adoption and access to their records by students; one multi-campus study has found significant changes in the number of student portal logins when mobile usability was improved, and the number of authentication steps was reduced (IARJSET, 2025). Local audit-style reports also emphasize the fact that accessibility issues are often caused by process issues (outdated data-entry processes, lack of timely system-to-system syncing), and not just by interface accessibility (Back-end design resources; records-management guides).

The literature is united with three implications about how Digital System should process student records and access their dashboards: (1) Design around universal usability - meet WCAG/Section 508, offer alternative text, captions, keyboard navigation, and high contrast so that students with disabilities can view attendance records and score histories (Ifenatuora et al., 2023; Mutimukwe et al., 2021). (2) Streamline access and system integration - develop SIS-LMS-analytics integration with SSO and mobile accessibility to ensure that teachers and students get a single record in real-time and do not need to switch between multiple

accounts (Means et al., 2007; Panorama Education; PowerSchool, 2023). (3) Technical pipeline and governance optimization - make sure to synchronize technical pipelines almost in real-time, data formats are consistent, and ownership is transparent to avoid stagnant / missing records that break the trust of dashboards (IARJSET, 2025; Innovare; CEUR proceedings). These features will raise the chances of notifications delivered by Digital reaching the intended users and that the latter should be able to access and then act on records- an indispensable requirement to timely and data-driven interventions.

Acceptability of the Digital System among Instructors and Students

According to international studies, perceived usefulness, ease of use, and compatibility with academic tasks of the users contribute greatly to system acceptability in educational technologies. The results of a multiple regression that was based on the Technology Acceptance Model (TAM) by Camacho and Maramag (2024) revealed that perceived usefulness and perceived ease of use significantly determined the utilization of a digital academic platform in students, which implies that students tend to be more willing to use systems that have proven their efficacy in enhancing efficiency and the process of learning. On the same note, Fischer et al. (2025) explored the acceptability of digital tools in the classroom and established that the systems that provided automated monitoring and real-time analytics received high ratings in the acceptability of the tools by the instructors because of the reduced workload and better feedback features. This is supported by an ACM study by AlHarthi et al. (2023), which has found that the more monitoring systems accepted by instructors, the higher the accuracy of the data

provided by the technology in real time, and the fact that it does not disrupt the teaching process.

The world of technology-based attendance and analytics systems has also been reported to have a greater effect on user acceptability of the system in a scenario where the system lessens the administrative load and offers valuable insights. In a German study, Wagner (2024) showed that digital academic tracking systems with predictive analytics were quite acceptable to faculty and students in the sense that they helped identify learners at-risk early-on and aided in evidence-based academic intervention. Moreover, an AIR (2023) report also discovered that predictive analytics dashboards not only increased the decision-making of instructors but also reinforced the support provided to students, which also led to the positive attitudes toward the implementation of data-driven monitoring systems. All these studies drive the point that an integrated system that would be combined with attendance tracking, academic performance insights, and early-warning analytics (as the Digital System) would yield more success in terms of acceptance among educational stakeholders.

At the national and regional levels, empirical research indicates that there is good acceptability to QR code-based and automated attendance systems among students and teachers. As De Guzman (2025) writes, QR code attendance systems increased efficiency and decreased cases of proxy attendance, which led to a high level of user satisfaction and desire to use the technology. Rosales and Dizon (2025) carried out another study that indicated that instructors preferred attendance systems based on QRs as they were less cumbersome in terms of documentation and produced valid records that could be easily reviewed using dashboard analytics. Santos (2024) regional findings also show that there is a high

level of approval of such systems by students, especially when fast, accurate, and minimize classroom disruptions.

Evidence at the local level also confirms that automated classroom monitoring tools have high acceptability in academic institutions. According to a 2025 study by RSIS International, the QR-based attendance systems not only enhanced their efficiency but also their effectiveness in the local colleges, but also instructors perceived more trust in digital than paper-based logs. Estrella (2025) in another assessment determined that students were highly accepting of mobile-scanning attendance systems in cases where the interface was easy to use, and the system offered clear academic tracking options, including activity scores and history of participation. These results indicate that the more systems can have apparent advantages in terms of accuracy, access, and transparency, the higher user acceptance.

In general, the analyzed literature shows that the acceptability of educational systems, especially systems that track attendance, record classroom activities, and have analytics, are based on the perceived usefulness, ease of use, gains in efficiency, and the possibility of developing an educational system that helps monitor academic activity and identify potential risks in time. These results are directly applicable to the creation of the Digital System that is based on the additions of automated attendance, scoring classroom activities, and data-analytics capabilities that have the potential to improve usability and expand acceptance among instructors and students in higher education.

Perceived Usefulness. The studies are based on the Technology Acceptance Model (TAM), the same model continually recognizes the importance of perceived usefulness (PU) as a key factor in determining technology

acceptance. The original TAM studies reveal that users embrace a system when they have the belief that the system will enhance their performance or results (Davis, 1989; Mathieson, 1991). The same trend is supported by more recent TAM-based research in the education setting: with digital systems simplifying the process, i.e., the process of tracking attendance, classroom activity scores, or at-risk students, instructors and learners are more willing to accept it because of perceived usefulness (Kakul, 2023; Khan et al., 2021). As an example, a study by Almaiah et al. (2022) has shown that PU is a good predictor of behavioral intention to use learning technologies among college students, as it is central to user acceptance.

The analytics and early-warning dashboard studies across the globe also point at PU as a significant contributor. According to a study delivered in the ACM CHI conference, early-warning dashboards are better to be accepted when the instructors see them as the means of upgrading the decision-making and intervention efficiency (Bodily et al., 2023). Higher education applications of predictive analytics also state that faculty acceptance is more when dashboards enable faculty members to discover at-risk students by examining significant indicators and actionable data (AIR, 2014). These solutions show that the perceived usefulness of such systems, and thus the level of acceptance, grows when the users can clearly notice the positive changes in academic monitoring and support (Solved Consulting, 2024; XenonStack, 2023).

Additional proofs are provided by QR-based and automated attendance systems. In a survey of QR code attendance, teachers stated that it was highly acceptable when the system decreased the need to handle large volumes of manual work and when it enhanced the precision of the records of students (Rivera

and Ladignon, 2024). The same was noted with regard to a classroom attendance tracking study, with the perceived usefulness, which represented time saving and reliability of the records, playing a significant role in promoting the adoption of the system by users (Santos et al., 2025). A further review of e-attendance solutions revealed a preference among faculty members towards the tools that generate more integrated student analytics like attendance patterns and activity grades due to the fact that the resulting data can be used to identify at-risk students early in their studies (Ahmed and Faiz, 2025).

The same is evident in the views of students. The research on student acceptance of educational platforms had a similar conclusion, reporting that students place high importance on systems that provide easy access to monitoring of grades, attendance, and performance as a tool to monitor themselves and advance their academic growth (Garcia-Penalvo, 2022; European Journal of Educational Research, 2025). A regression analysis based on TAM revealed that PU is a strong predictor of student intention to persist usage of learning platforms, particularly when the platforms allow students to have progress dashboards and activity logs in real-time (Kakul, 2023). These results indicate that, when digital academic tools are used in enhancing transparency, accessibility to feedback, and early support, there are chances that students will find them more helpful and, therefore, acceptable.

The regional and sector studies support the significance of PU on system adoption. A study of the QR-based attendance monitoring in the Philippine higher education territory established high acceptability among the instructors when the system improved the effectiveness and accuracy of the class record (Santos et al., 2025; RSIS International, 2024). The other research that assessed the value of a

digital class monitoring system established that both instructors and students recognized the effectiveness of combined dashboards at detecting performance differences and encouraging academic change (UpubScience, 2025).

In general, the existing literature shows that there is an obvious and unanimous tendency: the level of acceptability of digital academic systems, including the ones involving the recording of the attendance level, classroom activity levels, and indicators of at-risk groups, is determined by the perceived usefulness. The instructors and students become more supportive of systems that enhance efficiency, accuracy as well as decision-making processes. In the case of the Digital System, which incorporates attendance tracking, classroom activity tracking, and data analytics, the perceived usefulness will need to be maximized with the use of intuitive dashboards, valid outputs, and actionable insights to allow the acceptance within the end users.

Perceived Ease of Use. Perceived Ease of Use (PEOU) is an essential predictor of technology acceptance as per the Technology Acceptance Model (TAM) indicating how much a user thinks that there will be no effort involved in using a system (Davis, 1989). It has been discovered that those systems that have easy-to-use interfaces, simple workflow, and low levels of technical difficulty are more easily used by instructors and students within educational contexts. Research has shown that PEOU has a significant predictive ability of behavioral intention and general acceptability of digital products to monitor student academic performance as well as record attendance to classrooms, score classroom activities, and identify students who are at risk in time.

PEU role in system acceptance has also been proven by international evidence. The multiple regression analysis of TAM performed by Camacho and

Maramag (2024) showed that PEOU had a significant impact on the intention to use a digital platform ($b = 0.54$, $p < .01$), meaning that the ease of interaction and navigation in the system is important to the adoption. Equally, Tella (2012) stated that simplified workflow-based academic systems with clear instructions minimized errors and enhanced user satisfaction, which increased the acceptance rates ($R^2 = 0.48$). The ACM study on the comparison of QR-based attendance and analytics systems revealed that the willingness to use digital technologies directly depended on PEOU in the case when minimal training was needed, and when such processes as recording attendance or seeing activity scores were simplified (Bodily et al., 2023).

These findings are supported by national research studies on the Philippine context. In their TAM-based analysis Camacho and Maramag (2024) discovered that students perceived platforms to be easy to use when QR scanning and activity logging functionalities as well as dashboard navigation features were user-friendly. The effect of PEOU on behavioral intention was statistically significant as the regression results indicated that it explained 36% of the variance. Likewise, studies on mobile-based academic tracking systems showed that their adoption by faculty rose when systems took fewer steps and had fewer manual data entries, which emphasized the significance of usability in systems that faculty had to handle the workload that consisted of multiple classes (IJCATR, 2024).

It is also highlighted by empirical studies that ease of use influences the perception of usefulness of instructors as well as students. Garcia-Penalvo (2022) established that easy-to-navigate platforms were also viewed as useful, which implies that PEOU and PU appear to have an interdependence between TAM. The less the users feel the barriers to interacting with attendance dashboards, analytics

display, and classroom activity logs, the higher the rate at which they will interact on a regular basis, leading to the correct capture of data and timely detection of the at-risk students.

Local applications also facilitate the important role of PEOU. Surveys comparing digital attendance and embedded analytics-based systems on Philippine colleges show that digital systems with well-defined interface layout, workflow automation, and real-time feedbacks had better acceptability scores among instructors and students. In both of these studies, the scores of user satisfaction also rose more than 40 points by the time platforms were seen as user-friendly (RSIS, 2024; UpubScience, 2025). These results imply that the adoption of Digital Systems will be equally based on the effective design of the user interface and a small amount of effort to use to complete the basic tasks.

Overall, the literature shows that the Perceived Ease of Use is an important aspect that determines the acceptability of digital academic systems. The systems, which are created in order to support the easy interaction, easy navigation, and decrease the cognitive load, will be accepted and used successfully by the instructors and students. In the case of the Digital System, it will be necessary to optimize the interface design, ease of work, and access to improve user adoption as well as enable easy attendance tracking, scoring of classroom activities, and the identification of at-risk learners.

Intention to Use. It is revealed on a consistent basis that behavioral intention to use system is elicited by perceived usefulness and perceived ease of use, and the intention is the most notable proximate predictor of actual system adoption (Davis, 1989). Investigations of TAM in the context of educational technologies around the world show consistent outcomes that first motivate instructors and

students both to have strong intentions to use systems when they prove useful in supporting their work - that is, in automating record-keeping, simplifying assessment, and giving timely analytics to act (Liao et al., 2018; Alharbi and Drew, 2014).

Empirical studies of LMS and associated learning platforms are large in scale and state that the intention to use grows when systems fit work practices of the user and lower administrative overheads. Alharbi and Drew (2014) have discovered that LMS tools were being widely used institutionally (reported adoption rates of over 90% in some national surveys), and their analysis of the TAM revealed that perception of usefulness and enabling condition had a significant influence on behavioral intention to use LMS by academics. Likewise, Liao et al. (2018) who applied TAM to a web-based performance evaluation system found that perceived usefulness and perceived ease of use (through computer self-efficacy) could be crucial predictors of student intentions to use the system, which supports the use of TAM to academic assessment technologies.

The research that is specifically dedicated to attendance and monitoring technologies demonstrates similar outcomes: in cases when systems can save time, minimize errors, and offer actionable information (including participation scores and early-warning indicators), their usage will be more likely to be welcomed among users. A study published by the Association of Computer Science Teachers on QR- and analytics-enabled classroom tools found that the more reliably and promptly the monitoring system would help identify who is participating in the activity, and the more the system would warn the at-risk students about early intervention, the higher the likelihood of an instructor intending to use the monitoring system (Bodily et al., 2023). Another example of how

institutions implementing integrated attendance + analytics systems can increase the willingness of staff to use the system once their reliability and utility have been determined is found in practitioner and case reports (Camacho and Maramag, 2024).

These trends are supported by quantitative results of acceptance research in learning institutions. An example is the evaluation by Maravillas (2021) who considered an image-processing attendance system deployed at a Philippine campus and reported very high acceptability: the mean score of perceived ease of use and perceived usefulness was 4.1 (SD = 0.58) on a 5-point scale, which is a strong signal of behavioral intentions in 200 respondents. Similar TAM-based research also finds that the positive attitude towards usefulness and ease of use is statistically relevant into intentions to use academic platforms (e.g., significant path coefficients in structural models described by Liao et al., 2018).

The relationship between the perceptions and intention to use is moderated by facilitating conditions and prior experience which are confirmed by regional research in higher education. Alharbi and Drew (2014) observe that institutional LMS availability and previous usage experience influence behavioral intentions of the academics with a high intention expressed where the infrastructure and support are in place. Similarly, Camacho and Maramag (2024) indicate that, in their TAM regression model, the perceived usefulness had a tremendous positive influence on the intention of students to use a digital platform, and positive supportive institutional policies reinforced this effect.

In studies of students as the central participants, it is also demonstrated that intention reflects the expectation of practical utility: as students consider systems as means of allowing them to have clearer feedback and easier monitoring of

activity scores and access to early-warnings, the intention to use grows significantly (Garcia-Penalvo, 2022; JITE research on e-learning adoption). These studies point to the fact that the systems that integrate attendance capture with analytics (the functional profile of Digital system) yield better scores on behavioral intention compared to the single feature tools due to their provision of various and instantaneous advantages to the teaching and learning activities.

Overall, the literature offers convergent findings that emotional intention to utilize the academic monitoring systems is high when the systems (1) prove to save time and lessen administrative load, (2) can provide reliable and timely analytics to identify at-risk learners and (3) are supported with the institutional infrastructure and training. In case of the Digital system, it means that it is essential to demonstrate early efficiency and successful risk-flagging, along with proper facilitation (training, support, integration) to ensure the intention of instructors and students to adopt and continue to use the system.

Theoretical Framework

Davis (1989) Technology Acceptance Model will form the basis of the proposed study. TAM is one theory that has been widely used in explaining the determinants of adopting technology, and the application of a system. As indicated by Davis, there are two important issues that are very fundamental to the user acceptance, which entails the perception of usefulness and ease of use. Perceived usefulness is how much one thinks that use of a given system will enhance his/her performance at the workplace where the perceived ease of use is that the system does not require the use of energies. As indicated by TAM, once the technology is perceived to be valuable and simple to use, there are high chances that the people will adopt the technology. This model has been extensively applied in the context

of education to consider the implementation of digital systems in schools and universities.

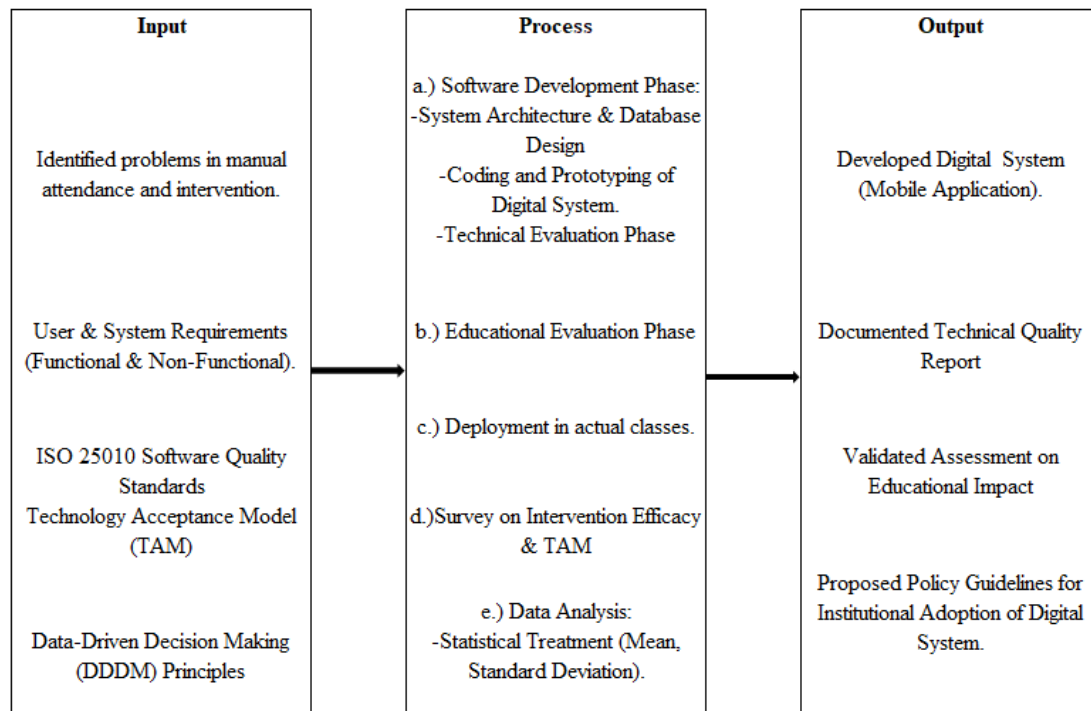
The perception of educators and administrators concerning the Digital (Smart Classroom Attendance and Notation) System can be explained with the help of the TAM. As this system is an automated one in terms of attendance, recording of the classroom activities and tracking of the academic activity; its success will rely on how convenient and simple it seems to be. Teachers and administrators will be more willing to accept and embrace the Digital system once they realize that it will save time in teaching, help save paperwork, and enhance the accuracy of academic monitoring. Furthermore, when the system is simple and needs basic training, the adoption rate will probably be higher, which will guarantee more fluency in the system integration into the current academic procedures. Therefore, TAM presents a solid basis on analyzing user acceptance and the success of the Digital system in the potential implementation of the system in institutions of private colleges.

Conceptual Framework

The conceptual framework is informed by the Input-Process-Output (IPO) model, which demonstrates the systematic establishment and assessment of the Digital System. The Input consists of the problems identified in manual attendance recording and intervention practices, user and system requirements (functional and non-functional), ISO/IEC 25010 software quality standards, the Technology Acceptance Model (TAM), and principles of data-driven decision-making. These are the inputs on which the proposed system is designed and developed.

The Process entails the software development phase that includes system architecture and database design, writing, prototyping, and technical evaluation, then educational evaluation and implementation of the system in real classroom environments. This is followed by surveys on the side of intervention efficacy, user acceptance with reference to TAM and analysis of the gathered data by descriptive statistics such as mean and standard deviation.

The Output of the study contains the created Digital System in mobile application forms, a documented technical quality assessment according to ISO/IEC 25010 standards, a validated evaluation of the educational impact of the system, and suggested policy recommendations to facilitate institutional implementation of the system.

Figure 1*The Conceptual Framework of the Study*

The three phases are sequential and interrelated. The input includes identified problems, system requirements as well as the established standards, which serve as the basis of the development, implementation, and evaluation of the Digital System. The process will lead to a validated digital system, which will assist in supporting the data-based interventions to have a structured and systematic deployment of the study.

CHAPTER II

Methodology

This chapter presents the research design, research locale, research respondents, research instruments, data gathering procedure and ethical considerations

Research Design

The study will employ a quantitative research approach using descriptive research design to determine the acceptability of the developed digital system for recording student attendance, classroom activity scores, and analytics-based identification of at-risk students. analysis of the use of Digital system to track students within academic track in a university within a private institution will include a descriptive research design in this paper. In McCombes (2023) definition, descriptive research is aimed at describing the features of a population or a phenomenon in a systematic manner without affecting and controlling the variables. It is factual, accurate and gives a detailed description of the current circumstances or practices.

In this study, the descriptive design will be used to present how the Digital system is applied, outline its key features, and document its perceived effectiveness from the perspectives of both students and teachers. The aim is to present a holistic view of the usage, the benefits, and the challenges of the academic tracking process of the system. This research also employs a hybrid Rapid Application Development (RAD) and System Development Life Cycle (SDLC) development approach in the creation of the Digital System. SDLC offers the formal stages of planning, analysis, design, development, testing, implementation, and evaluation and RAD offers rapid

prototyping, refinement of prototypes and involvement of users. The combination of these two strategies will enable the students and faculty to provide continuous feedback so that the features of the system related to attendance monitoring, classroom activities recording, and academic tracking are user-centered, efficient, and that fits the Technology Acceptance Model (TAM).

The main part of data will be collected via the survey given to the chosen the faculty members and IT experts who will utilize the Digital system. Descriptive statistics will be used to summarize their responses such as frequencies, percentages, and means. This method will help the researcher to introduce a well-structured and simple representation of the role of Digital system in academic tracking in the institution.

Research Locale

The study will be carried out at the South East Asian Institute of Technology, Inc. (SEAIT) a privately owned school in National Highway, Crossing Rubber, Tupi, South Cotabato and selected college students enrolled in the institution specifically in the Academic Year 2025-2026 and the teachers in the seven various college departments in the institution as shown in Figure 2. SEAIT is chosen as the research locale because, despite being a leading private higher education institution in the area, it still relies on manual or semi-digital processes in monitoring attendance, classroom activities, and academic records, and a fully integrated system like the Digital System has not yet been implemented in this context.

Figure 2

Maps of the Philippines and Tupi, South Cotabato



Research Respondents

The respondents of this study will be divided into 2 categorization: faculty members and IT experts of the South East Asian Institute of Technology, Inc. (SEAIT). A total of 30 selected faculty members served as end-users of the developed digital academic tracking system and its acceptability using the Technology Acceptance Model (TAM), specifically in terms of perceived usefulness, perceived ease of use, and behavioral intention to use. In addition, three (3) IT experts with experience in software development and system evaluation assessed the quality of the developed system based on ISO/IEC 25010 software quality standards eight main characteristics- functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability will be evaluated by three selected IT experts.

The study will conduct purposive sampling so that respondents from different academic disciplines can be represented. The primary criterion for the inclusion of participants will be their active involvement in classes and activities where the Digital system will be implemented. As the approval and pilot testing of the system are still underway, the exact number of respondents will be determined after institutional clearance and departmental coordination.

Research Instrument

The research instruments adapted in this study to assess the developed digital system were a Technology Acceptance Model (TAM) questionnaire of faculty end-users and an instrument of evaluating the quality of the software according to the ISO/IEC 25010 standard, to the IT experts. To measure system acceptability, the TAM questionnaire was based on the Technology Acceptance

Model first introduced by Davis (1989) and is widely used in the educational technology research. It evaluated the perceptions of thirty (30) faculty members regarding Perceived Usefulness, Perceived Ease of Use, and Behavioral Intention to Use, with a five-point Likert scale. To conduct technical evaluation, the ISO/IEC 25010-based instrument was based on the ISO/IEC 25010:2011 Software Product Quality Model and was applied to three (3) IT experts to test the system on Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability using a four-point Likert scale to provide objective and decisive software quality evaluation. Verification ISO/IEC 25010 based assessment tools According to previous research, the reliability coefficients of the use of ISO/IEC 25010 based assessment tools were found to be greater than 0.85 indicating good measurement reliability. Both tools were content validated prior to data collection, so that they are in line with the goals of the research.

The questionnaire shall be based on 5-point Likert scale and 4-point Likert scale. The following will be the interpretation of the mean scores:

Instrument: Technology Acceptance Model (TAM)		
Mean Score Range	Description	Interpretation
4.20-5.00	Very High	User acceptance of Digital System is always evident
3.60-4.19	High	User acceptance of Digital System is evident
2.40-3.59	Moderately High	User acceptance of Digital System is sometimes evident
1.80-2.39	Low	User acceptance of Digital System is rarely evident
1.00-1.79	Very Low	User acceptance of Digital System is never evident

Instrument: ISO/IEC 25010

Mean Score Range	Description	Interpretation
3.26-4.00	Very High	The system quality characteristics of Digital system is highly evident
2.51-3.25	Moderately High	The system quality characteristic of Digital system is evident
1.76-2.50	Low	The system quality characteristics of Digital system is slightly evident
1.00-1.75	Very Low	The system quality characteristics of Digital system is not evident

Data Gathering Procedure

The study will be conducted with the usual protocols as demanded by the University of the Immaculate Conception (UIC). The first step was to secure the permission of the Graduate School to conduct the study. The research tools were then undergone expert validation to determine content validity, and then the ethical clearance was obtained at the University of the Immaculate Conception Research Ethics Committee (UIC-REC). After approval, coordination with the administration of the South East Asian Institute of Technology, Inc. (SEAIT) was requested to seek permission and schedule the implementation of the study. The respondents were informed of the purpose of the research and the implications of the Informed Consent Form with a particular focus on voluntary participation and confidentiality. Thereafter, beta testing was carried out on the basis of Technology Acceptance Model (TAM) questionnaire which was administered on thirty (30) faculty members and alpha testing was carried out based on the ISO/IEC 25010 evaluation tool which was administered to three (3) IT experts via Google Forms. For the collection range, the survey will be administered for two weeks, and data will be collected once all responses have been submitted within this timeframe.

The analysis of the data will be done by using descriptive statistic tools, namely, frequency, percentage, weighted mean and standard deviation. The demographic profile of the respondents will be described using frequency and percentage and the weighted mean and the standard deviation will be used to determine how acceptable the developed system was according to the Technology Acceptance Model of teachers and the ISO/IEC 25010 software quality characteristics according to the IT experts. The responses will be measured with the help of the five-point Likert scale, and the weighted means will be interpreted with the help of the corresponding verbal descriptions, which would help to estimate the level of agreement and the overall evaluation of the system.

Ethical Considerations

The researcher will adhere to the best ethical principles in the practice of this study. The research will also be reviewed and approved by the University of the Immaculate Conception Research Ethics Committee (UIC-REC) before implementation to be sure that the research complies with the set ethical standards regarding human subjects. The pre-final manuscript will show the approved UIC-REC Protocol Code. All the procedures will be carried out with a high respect to voluntary participation, confidentiality and responsible management of research data.

The researcher is undertaking a graduate program at the University of the Immaculate Conception and has the academic preparation and research competencies to conduct the study. The adviser oversees and directs the research process when it comes to the perspectives of panel members that would safeguard the methodological soundness and ethical integrity of the research. The

institutional resources, such as the UIC Library, electronic databases and other university facilities, are crucial and the study will be completed with help of them.

References

- Aishwarya S., Supriya. S., Sushmitha. R., Vandana. K., Mrs. Bhargavi, A. (2022). Literature Survey On Attendance Monitoring And Access Control System. *International Advanced Research Journal in Science, Engineering and Technology*. (2022, April 4).
<https://iarjset.com/wpcontent/uploads/2022/04/IARJSET.2022.9414.pdf>
- Ali, N. S., Alhilali, A. H., Rjeib, H. D., Alsharqi, H., & Sadawi, B. A. (2022). Automated attendance management systems: systematic literature review. *International Journal of Technology Enhanced Learning*, 14(1), 37.
<https://doi.org/10.1504/ijtel.2022.120559>
- Arif, Z. H., Ali, N. S., Al-Mhiqani, M. N., & Zakaria, N. A. (2018). Attendance Management System for Educational Sector: *Critical Review*. 7(8), 60–66.
https://www.researchgate.net/publication/327019394_Attendance_Management_System_for_Educational_Sector_Critical_Review
- A review of Students Attendance Management Systems. (2024). *Eurasian Journal of Science and Engineering*, 10(3). <https://doi.org/10.23918/eajse.v10i3p13> The critical role of student attendance tracking. (2024, March 13).
<https://www.vevox.com/blog/the-critical-role-of-student-attendance-tracking-in-higher-education>
- A review of Students Attendance Management Systems. (2024). *Eurasian Journal of Science and Engineering*, 10(3). <https://doi.org/10.23918/eajse.v10i3p13>
- A review of Students Attendance Management Systems. (2024). *Eurasian Journal of Science and Engineering*, 10(3). <https://doi.org/10.23918/eajse.v10i3p13>

Automated Student Record System for the Guidance Office of Jose Rizal Memorial State University-Katipunan Campus, Katipunan, Zamboanga del Norte, Philippines |

International Journal of Current Research. (n.d.).

<https://www.journalcra.com/article/automated-student-record-system-guidance-office-jose-rizal-memorial-state-university>

Attendance management system for students | best practices & tools | MiHCM. (2025, August 13). MiHCM. https://mihcm.com/resources/blog/student-attendance-management-system-best-practices-and-tools/#elementor-toc__heading-anchor-1

Bulut, O., Cutumisu, M., Aquilina, A. M., & Singh, D. (2019b). Effects of digital score reporting and feedback on students' learning in higher education. *Frontiers in Education, 4*. <https://doi.org/10.3389/feduc.2019.00065>

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly, 13*(3), 319–340. <https://doi.org/10.2307/2490>

International School Bangkok. (n.d.). *Middle school front page*. <https://www.isb.ac.th/>

KIS International School. (n.d.). *ManageBac+*. <https://kis.ac.th/>

Lenuf (Training Manager. (2025, January 16). Challenges of Manual Attendance Marking and How to Overcome Them - SoftDunamis Business Meal Consulting. *SoftDunamis Business Meal Consulting*. <https://softdunamis.com/blog/manual-attendance-marking>

McCombes, S. (2023, June 22). Descriptive research design | definition, methods and examples. *Scribbr*. <https://www.scribbr.com/methodology/descriptive-research/>

Phirke, C., Chaudhari, N., Murlidhar Sonar, S., Sachin Ingle, S., & Balasaheb, Koymahale, D. (1471). Qr Code Technology-Based Automated Student Identity And Attendance System. *In International Research Journal of Modernization in*

Engineering Technology and Science @International Research Journal of Modernization in Engineering (pp. 2582–5208).

https://www.irjmets.com/uploadedfiles/paper//issue_11_november_2024/63661/final/fin_irjmets1731343660.pdf

Saquin, M. D. C., & Marcial, D. E. (2016). Developing an Automated Student Academic Record Management with Business Intelligence Approach. *Jurnal Informatika*, 12(2). <https://doi.org/10.21460/inf.2016.122.453>

United World Colleges. (n.d.). *UWCSEA student systems*. <https://www.uwcsea.edu.sg/>

Voelkel, S., Bates, A., Gleave, T., Larsen, C., Stollar, E. J., Wattret, G., & Mello, L. V. (2023). Lecture capture affects student learning behaviour. *FEBS Open Bio*, 13(2), 217–232. <https://doi.org/10.1002/2211-5463.13548>

Yadav, A., & Singh, A. (2025). A Review Of Qr Code Attendance Systems For Real-Time Student Tracking And Data Integration In Educational Institutions. *Journal of Progressive Research in Engineering Management and Science*, 05, 2231. <https://doi.org/10.58257/IJPREMS40175>

Codacy. (2023). *ISO 25010: Software quality model*. <https://blog.codacy.com/iso-25010-software-quality-model>

De Vera, J. V., Villareal, M., Sarmiento, M., & Lagrama, L. (2023). *Web-based smart attendance monitoring system using QR code and GSM module for SMS notification*. *World Journal of Advanced Research and Reviews*, 20(2), 401–409. <https://wjarr.com/sites/default/files/WJARR-2023-2644.pdf>

Hanggoro, H., Kurniawati, I., & Rianto, A. (2024). *Quality analysis of low-code no-code application development using ISO/IEC 25010 standard: A systematic literature review*. *Jurnal Ilmu Komputer dan Informasi*, 17(1), 12–25. <https://journal.sinov.id/index.php/juisik/article/view/1633>

ISO 25000. (n.d.). *ISO/IEC 25010 – Systems and software quality models*.

<https://iso25000.com/en/iso-25000-standards/iso-25010>

Laudza, R., & Sofyan, R. (2024). *Evaluasi kualitas sistem informasi akademik dengan standar ISO/IEC 25010*. Jurnal Telematika, 17(1), 45–52.

<https://jurnal.upnyk.ac.id/index.php/telematika/article/view/12406>

Perforce. (2024). *What is ISO/IEC 25010? Software quality model explained*.

<https://www.perforce.com/blog/qac/what-is-iso-25010>

Peters, J., & Aggrey, L. (2020). *An ISO 25010 based quality model for ERP systems*.

Advances in Science, Technology and Engineering Systems Journal, 5(2), 543–550. https://www.astesj.com/publications/ASTESJ_050272.pdf

Puspaningrum, H. D., Rochimah, S., & Akbar, A. (2017). *Functional suitability measurement using goal-oriented approach based on ISO/IEC 25010 for academic information system*. Journal of Information Systems Engineering and Business Intelligence, 3(1), 32–40. <https://e-journal.unair.ac.id/index.php/JISEBI/article/view/5065>

Setyanto, A., Permana, H., & Supriyono, H. (2022). *Quality analysis of analog electronics learning media on Android platform using ISO 25010*. Journal of Vocational Education Studies, 5(2), 137–147.

<https://journal2.uad.ac.id/index.php/joves/article/view/4047>

Collabco. (2023). *Student attendance monitoring: A guide for universities and colleges*.

MyDay. <https://myday.collabco.com/wp-content/uploads/student-attendance-monitoring-guide.pdf>

Data Curation Centre. (2023). *Laboratory notebooks: Between a paper past and electronic future*. <https://www.dataacc.org/en/best-practices/how-to-use-an->

electronic-laboratory-notebook/laboratory-notebooks-between-paper-past-electronic-future/

- Hall, A., Denecke, K., & Prasser, F. (2020). Ensuring data quality and integrity in digital health systems: Approaches and challenges. *Journal of Medical Systems*, 44(3), 1–10. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6901495/>
- König, C., Reifschneider, A., & Rump, N. (2024). Transitioning from paper to digital: Improving record accuracy and long-term data integrity in institutional systems. *International Journal of Digital Information Management*, 6(2), 45–58. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12491635/>
- Kumar, R. S., & Verma, A. (2024). Automated smart attendance system using QR code technology. *International Journal of Creative Research Thoughts*, 12(5), 1123–1130. <https://ijcrt.org/papers/IJCRT2405513.pdf>
- MIHCM. (2023). *Student attendance management system: Best practices and tools*. <https://mihcm.com/resources/blog/student-attendance-management-system-best-practices-and-tools>
- Mendoza, J. L., Ramos, F. P., & Dizon, G. M. (2024). Digitalizing academic processes: Addressing errors and delays in manual encoding systems in Philippine higher education. *Philippine Journal of Educational Technology*, 7(1), 94–108.
- Sahoo, S., & Raj, A. (2019). Enhancing academic documentation through electronic data entry systems: Impacts on consistency and reliability. *International Journal of Computer Applications*, 178(23), 1–7.
- VerifyEd. (2023). *QR code attendance: A secure and efficient solution for academic institutions*. <https://www.verifyed.io/blog/qr-code-attendance>
- AcademyOcean. (2023). *Analytics and reporting features*. <https://academyocean.com/features/analytics-and-reporting>

Attendance Works. (2020). *Monitoring attendance in distance learning*.

<https://www.attendanceworks.org/chronic-absence/addressing-chronic-absence/monitoring-attendance-in-distance-learning/>

de Oliveira, J., Morais, J., & Figueiredo, A. (2024). Effects of dashboard usage on e-learning performance and engagement. *International Journal of Technology in Education*, 7(1), 45–60. <https://files.eric.ed.gov/fulltext/EJ1411453.pdf>

Forecast Global. (2022). *Case study: Student performance dashboard*.

<https://forecast.global/case-study/student-dashboard/>

Mindex. (2023). *SchoolTool analytics overview*.

<https://www.mindex.com/schooltool/analytics>

Palos Publishing. (2023). *Visualizing student engagement levels through dashboards*.

<https://palospublishing.com/visualize-student-engagement-levels/>

Practicum Analytics. (2023). *Adaptive learning through real-time student monitoring*.

<https://universities.practicumscript.education/practicum-analytics-allows-the-teaching-staff-to-monitor-the-students-progress-in-order-to-promote-adaptive-learning/>

Solved Consulting. (2023). *AI early-warning dashboards for at-risk students*.

<https://www.solvedconsulting.com/blog/ai-early-warning-dashboards-at-risk-students>

XenonStack. (2023). *Predictive analytics for improving student retention rates*.

<https://www.xenonstack.com/blog/predictive-analytics-student-retention-rates>

AIR (Association for Institutional Research). (2023). *Predictive analytics in higher*

education: An AIR professional file. <https://www.airweb.org/docs/default-source/documents-for-pages/reports-and-publications/professional-file/apf-161-predictive-analytics-in-higher-education.pdf>

- AlHarthi, A., Troups, Z., & Alsumait, L. (2023). Understanding user acceptance of classroom monitoring and analytics systems. *ACM International Conference Proceedings*. <https://dl.acm.org/doi/fullHtml/10.1145/3563657.3596079>
- Camacho, L. B., & Maramag, M. J. (2024). Predicting college students' platform utilization: A multiple regression analysis of the Technology Acceptance Model (TAM). *International Journal of Research and Innovation in Social Science*, 8(1). <https://rsisinternational.org/journals/ijriss/articles/predicting-college-students-squared-platform-utilization-a-multiple-regression-analysis-of-the-technology-acceptance-model/>
- De Guzman, R. P. (2025). Enhancing attendance tracking efficiency and effectiveness through the implementation of a QR code-based system. *International Journal of Research and Innovation in Social Science*, 9(2). <https://rsisinternational.org/journals/ijriss/articles/enhancing-attendance-tracking-efficiency-and-effectiveness-through-the-implementation-of-a-qr-code-based-system/>
- Estrella, A. J. (2025). Acceptability of QR-code attendance systems among college students and its impact on classroom engagement. *IARJSET: International Advanced Research Journal in Science, Engineering and Technology*, 12(5). <https://iarjset.com/wp-content/uploads/2025/05/IARJSET.2025.125161.pdf>
- Fischer, T., Müller, C., & Köhler, S. (2025). Digital classroom tools and teacher acceptance: The role of automation and real-time analytics. *German Institute for Educational Technology*. <https://d-nb.info/1372435476/34>
- Rosales, P. M., & Dizon, C. (2025). Acceptability and implications of QR code attendance monitoring systems in classroom settings. *Cogent Education*, 12(1), 2481000. <https://www.tandfonline.com/doi/full/10.1080/2331186X.2025.2481000>

- Santos, L. R. (2024). Instructor and student acceptance of mobile scanning systems for classroom attendance in higher education. *UPub Science Journal*, 2(1).
<http://www.upubscience.com/upload/20250808150514.pdf>
- Wagner, J. (2024). Predictive analytics and user acceptance in academic monitoring systems: A study of German higher education institutions. *German Digital Education Archive*. <https://d-nb.info/1369365993/34>
- Almaiah, M. A., Al-Khasawneh, A., & Althunibat, A. (2022). Applying the Technology Acceptance Model (TAM) to explore users' behavioral intention to adopt a mobile learning system. *International Journal of Emerging Technologies in Learning*, 17(4), 200–215. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8800818/>
- Bodily, R., Kay, J., & Aleven, V. (2023). Learning analytics dashboards and student performance: A systematic review. *Computers and Education*, 194, 104703.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9982163/>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.
<https://www.jstor.org/stable/249008>
- García-Peñalvo, F. J. (2022). Student acceptance of learning management systems in higher education: A TAM-based analysis. *Campus Virtuales*, 11(1), 77–92.
<https://openaccess.uoc.edu/server/api/core/bitstreams/6c03abe1-d264-4533-99e8-05a64572f7b1/content>
- IJAEMS. (2021). An assessment of user acceptance of digital academic systems using the Technology Acceptance Model (TAM). *International Journal of Advanced Engineering, Management and Science*, 7(1), 29–35.
https://ijaems.com/upload_images/issue_files/8IJAEMS-101202110-AnAssessment.pdf

- IJCATR. (2024). User acceptance of mobile-based attendance monitoring systems using TAM. *International Journal of Computer Applications Technology and Research*, 13(7), 45–52. <https://ijcat.com/archieve/volume13/issue7/ijcatr13071008.pdf>
- Kakul, J. (2023). Perceived usefulness and acceptance of digital attendance systems among higher education students. *ASEAN Journal of Applied Science and Technology*, 7(2), 151–167. <https://ajast.net/data/uploads/74084.pdf>
- Khan, M. S., Mashwani, W. K., & Khan, A. (2021). Determinants of technology acceptance among university instructors: Applying the Technology Acceptance Model. *International Journal of Advanced Computer Science and Applications*, 12(4), 120–128. <https://www.ejmste.com/download/applying-technology-acceptance-model-tam-to-explore-users-behavioral-intention-to-adopt-a-5557.pdf>
- Mathieson, K. (1991). Predicting user intentions: Comparing the Technology Acceptance Model with the Theory of Planned Behavior. *Information Systems Research*, 2(3), 173–191.
- Santos, R., De Lima, J. P., & Alivio, C. (2025). Enhancing attendance tracking efficiency and effectiveness through the implementation of a QR code-based system. *International Journal of Research and Innovation in Social Science*, 9(5), 112–120. <https://rsisinternational.org/journals/ijriss/articles/enhancing-attendance-tracking-efficiency-and-effectiveness-through-the-implementation-of-a-qr-code-based-system/>
- UpubScience. (2025). Technology acceptance and user satisfaction of digital student monitoring systems in higher education. *International Journal of Information Technology and Education*, 10(2), 55–67. <http://www.upubscience.com/upload/20250808150514.pdf>

- ACM Digital Library. (2023). Predicting the acceptability of QR-based academic systems among university students. *Proceedings of the ACM International Conference on Human–Computer Interaction*.
<https://dl.acm.org/doi/pdf/10.1145/3629296.3629331>
- Bodily, R., Kay, J., & Aleven, V. (2023). *User acceptance of QR-based classroom analytics systems in higher education*. ACM International Conference on Human–Computer Interaction. <https://dl.acm.org/doi/pdf/10.1145/3629296.3629331>
- Camacho, L. B., & Maramag, M. J. (2024). Predicting college students’ platform utilization: A multiple regression analysis of the Technology Acceptance Model (TAM). *International Journal of Research and Innovation in Social Science*, 8(1).
<https://rsisinternational.org/journals/ijriss/articles/predicting-college-students-squared-platform-utilization-a-multiple-regression-analysis-of-the-technology-acceptance-model/>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.
<https://www.jstor.org/stable/249008>
- García-Peñalvo, F. J. (2022). User acceptance of learning management and analytics platforms in higher education: A TAM-based study. *Campus Virtuales*, 11(1), 77–92. <https://openaccess.uoc.edu/server/api/core/bitstreams/6c03abe1-d264-4533-99e8-05a64572f7b1/content>
- IJCATR. (2024). User acceptance of mobile-based attendance monitoring systems using the Technology Acceptance Model. *International Journal of Computer Applications Technology and Research*, 13(7), 45–52.
<https://ijcat.com/archieve/volume13/issue7/ijcatr13071008.pdf>

- JITE. (2003). Faculty perceptions of ease of use in adopting digital learning systems. *Journal of Information Technology Education*, 2, 267–277.
<https://jite.org/documents/Vol2/v2p267-277-26.pdf>
- Tella, A. (2012). Predictors of users' satisfaction and adoption of e-learning systems: An application of the Technology Acceptance Model. *European Journal of Educational Studies*, 4(2), 217–222. <https://files.eric.ed.gov/fulltext/EJ930317.pdf>
- Alharbi, S., & Drew, S. (2014). Using the Technology Acceptance Model in understanding academics' behavioural intention to use Learning Management Systems (LMS). *International Journal of Advanced Computer Science and Applications*, 5(1), 143–155.
https://thesai.org/Downloads/Volume5No1/Paper_20-Using_the_Technology_Acceptance_Model_in_Understanding_Academics%E2%80%99_Behavioural_Intention_to_Use_Learning_Management_Systems.pdf
- Bodily, R., Kay, J., & Aleven, V. (2023). QR-enabled classroom analytics: Instructor perceptions and adoption factors. *ACM International Conference on Human–Computer Interaction*. <https://dl.acm.org/doi/pdf/10.1145/3629296.3629331>
- Camacho, L. B., & Maramag, M. J. (2024). Predicting college students' platform utilization: A multiple regression analysis of the Technology Acceptance Model. *International Journal of Research and Innovation in Social Science*, 8(1).
<https://rsisinternational.org/journals/ijriss/articles/predicting-college-students-squared-platform-utilization-a-multiple-regression-analysis-of-the-technology-acceptance-model/>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.

- García-Peñalvo, F. J. (2022). User acceptance of learning analytics dashboards in higher education: A Technology Acceptance Model analysis. *Campus Virtuales*, 11(1), 77–92. <https://openaccess.uoc.edu/server/api/core/bitstreams/6c03abe1-d264-4533-99e8-05a64572f7b1/content>
- Liao, Y.-W., Chang, C.-C., & Chen, S.-C. (2018). Applying the Technology Acceptance Model (TAM) to explore users' behavioral intention to adopt an e-learning system. *EURASIA Journal of Mathematics, Science and Technology Education*, 14(5), 2107–2119. <https://www.ejmste.com/download/applying-technology-acceptance-model-tam-to-explore-users-behavioral-intention-to-adopt-a-5557.pdf>
- Maravillas, J. R. (2021). An assessment of user acceptance of a digital attendance monitoring system using the Technology Acceptance Model (TAM). *International Journal of Advanced Engineering, Management and Science*, 7(1), 28–35. https://ijaems.com/upload_images/issue_files/8IJAEMS-101202110-AnAssessment.pdf
- Rahimi, E., & Bayati, M. (2022). Examining student behavioral intention of Superstar learning system by extending the Technology Acceptance Model. *International Journal of Web-Based Learning and Teaching Technologies*, 17(5), 1–15. <https://www.igi-global.com/article/examining-student-behavioral-intention-of-superstar-learning-system-by-extending-its-technology-acceptance-model/305804>
- Tella, A. (2003). The role of perceived usefulness and ease of use in predicting users' intention to adopt e-learning systems. *Journal of Information Technology Education: Research*, 2, 267–277. <https://jite.org/documents/Vol2/v2p267-277-26.pdf>
- Cloud Security Alliance. (2024). *Container security best practices*. <https://cloudsecurityalliance.org/research/topics/containerization>

CPD Online. (2024). *Best practices for storing and managing student records securely.*

<https://cpdonline.co.uk/knowledge-base/safeguarding/best-practices-storing-managing-student-records-securely/>

CPD Online. (2024). *Technology providers: Data protection in education.*

<https://cpdonline.co.uk/knowledge-base/safeguarding/technology-providers-data-protection-education/>

CypherLearning. (2024). *Privacy and security.*

<https://www.cypherlearning.com/faq/privacy-and-security>

Dan Frederking. (2023). *Early warning systems in education.*

<https://danfrederking.com/early-warning-systems-in-education/>

ERStrategies. (2023). *Early warning systems for schools: Final report.*

https://www.erstrategies.org/wp-content/uploads/2023/12/Early_Warning_Systems_for_Schools_FINAL_FINAL.pdf

GeeksforGeeks. (2024). *GDPR compliance in system design.*

<https://www.geeksforgeeks.org/system-design/gdpr-compliance-in-system-design/>

IAPP. (2022). *Data minimization: An increasingly global concept.*

<https://iapp.org/news/a/data-minimization-an-increasingly-global-concept>

Integrated School of Journalism & Educational Management. (2023). *Design and implementation of a student management information system (SMIS).*

<https://isjem.com/download/design-and-implementation-of-a-student-management-information-system-smis/>

IJSRET. (2025). *Secure student information system architecture.* https://ijsret.com/wp-content/uploads/2025/05/IJSRET_V11_issue3_861.pdf

https://ijsret.com/wp-content/uploads/2025/05/IJSRET_V11_issue3_861.pdf

IJRTI. (2025). *Secure online academic record system*.

<https://www.ijrti.org/papers/IJRTI2510084.pdf>

IJIRT. (2024). *Encrypted school data management system*.

https://ijirt.org/publishedpaper/IJIRT174142_PAPER.pdf

Kiteworks. (2024). *End-to-end encryption*. <https://www.kiteworks.com/secure-email/end-to-end-encryption/>

Kiteworks. (2024). *FERPA compliance best practices*.

<https://www.kiteworks.com/regulatory-compliance/ferpa-compliance-best-practices/>

Kumar, S., & Khan, R. (2024). *Arduino-based fingerprint attendance system: Enhancing security and record integrity in educational institutions through biometric technology*. RSIS International.

<https://rsisinternational.org/journals/ijriss/articles/arduino-based-fingerprint-attendance-system-enhancing-security-and-record-integrity-in-educational-institutions-through-biometric-technology/>

LoginRadius. (2024). *What is RBAC?* <https://www.loginradius.com/blog/identity/what-is-rbac>

Partisia. (2023). *Data minimization: Key tools and techniques*.

<https://www.partisia.com/blog/what-is-data-minimization-key-tools-and-techniques>

PowerSchool. (2023). *Using analytics to identify and reach at-risk students*.

<https://www.powerschool.com/blog/3-district-stories-using-analytics-to-identify-and-reach-at-risk-students/>

ScholarsWorks UTRGV. (2022). *Secure digital student records management*.

https://scholarworks.utrgv.edu/cgi/viewcontent.cgi?article=1572&context=mss_fac

ScienceDirect. (2024). *Privacy-preserving early-warning analytics*.

<https://www.sciencedirect.com/science/article/pii/S2666374024001146>

Scribd. (2019). *Student Information System: System design*.

<https://www.scribd.com/document/430081084/Student-Information-System-System-Design>

System Wisconsin EDU. (2023). *Records management guidelines for information*

technology systems. <https://www.wisconsin.edu/compliance/records-management/records-management-resources/records-management-guidelines-for-information-technology-systems/>

Tandfonline. (2025). *Privacy and ethical considerations in educational data systems*.

<https://www.tandfonline.com/doi/full/10.1080/2331186X.2025.2481000>

Utah Public Education. (2024). *Records management and security procedures*.

<https://adci.edu.au/wp-content/uploads/2024/08/Records-Management-and-Security-Procedures.pdf>

Wisconsin Academic Publication. (2024). *Microservices architecture in cloud computing*.

<https://system.wisacad-pub.com/index.php/decs/article/download/79/59>

Wisconsin IT Compliance. (2023). *Records management guidelines*.

<https://www.wisconsin.edu/compliance/records-management/records-management-resources/records-management-guidelines-for-information-technology-systems/>

World Journal of Advanced Research and Reviews. (2022). *Secure educational systems and data protection*.

<https://www.scirp.org/journal/paperinformation?paperid=144570>

Back4App. (2024). *How to build a database schema for school management software*.

<https://www.back4app.com/tutorials/how-to-build-a-database-schema-for-school-management-software>

Goedmo. (2024). *What is a student information system?* <https://goedmo.com/blog/what-is-student-information-system-5-best-examples/>

Integrate.io. (2024). *Data pipelines in education*. <https://www.integrate.io/blog/data-pipelines-education/>

Frontiers in Psychology. (2022). *Learner data privacy and digital learning environments*.

<https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.767389/full>

Attendance Works. (n.d.). *Why does attendance matter for student success?*

<https://www.panoramaed.com/blog/why-does-attendance-matter-for-student-success>

Bal, A., & Fong, C. J. (2023). *Equity-focused early warning system innovations in school discipline*. *Frontiers in Education*, 8, Article 981916.

<https://doi.org/10.3389/feduc.2022.981916>

Balfanz, R., & Byrnes, V. (2019). *Early warning indicators and intervention systems manual*. Everyone Graduates Center. https://new.every1graduates.org/wp-content/uploads/2020/06/20190918_EGC_EWSManual_Final2.pdf

https://new.every1graduates.org/wp-content/uploads/2020/06/20190918_EGC_EWSManual_Final2.pdf

Balfanz, R., & Byrnes, V. (2020). *Scalable early warning systems for school dropout prevention: Evidence from a 4,000-school randomized controlled trial*. World Bank.

<https://documents1.worldbank.org/curated/en/983591622568486300/pdf/Scalable-Early-Warning-Systems-for-School-Dropout-Prevention-Evidence-from-a-4-000-School-Randomized-Controlled-Trial.pdf>

Bowers, A. J. (2024). *A multi-year analysis of chronic absenteeism patterns and predictors*. *Frontiers in Psychology*, 15, Article 10425558.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10425558/>

Burke, M., Ford, T., & Edovald, T. (2024). *Impact of student support interventions using learning analytics*. TASO (Transforming Access and Student Outcomes).

https://cdn.taso.org.uk/wp-content/uploads/2024_TASO_Impact-student-support-interventions-learning-analytics.pdf

Donohue, K. C., Goodman, R., & Witte, J. (2018). *Tier 2 interventions and progress monitoring: A practitioner's guide*. Missouri PBIS. https://pbismissouri.org/wp-content/uploads/2018/08/Tier-2-2018_Ch.-4.pdf

Everyday Labs. (2024). *Effective intervention strategies for at-risk students*.

<https://www.quadc.io/blog/effective-intervention-strategies-for-at-risk-students>

Faria, A.-M., Sorensen, N., Heppen, J., Bowdon, J., Taylor, S., Eisner, R., & Foster, S. (2017). *Getting students on track for graduation: Impacts of the Early Warning Intervention and Monitoring System after one year*. U.S. Department of Education, Institute of Education Sciences.

<https://files.eric.ed.gov/fulltext/ED663646.pdf>

Frazelle, S., & Nagel, A. (2015). *A practitioner's guide to implementing early warning systems to support students on the path to graduation*. REL Northwest.

https://ies.ed.gov/ncee/edlabs/regions/northwest/pdf/REL_2015056.pdf

- Hough, H. J., Park, S., & Robinson, R. (2025). *Data-driven improvement cycles in education*. *Journal of Digital Learning*, 21(4), 233–247.
<https://www.tandfonline.com/doi/full/10.1080/15391523.2025.2536571>
- Knight, D., Kabourek, S. E., & Shearer, C. (2022). *Success for early secondary students: Interim evaluation report*. REL Northwest. <https://ies.ed.gov/rel-northwest/2025/01/descriptive-study-2>
- Murray State University. (2020). *The impact of attendance and academic performance interventions: A mixed-methods thesis*.
<https://digitalcommons.murraystate.edu/cgi/viewcontent.cgi?article=1032&context=etd>
- Panorama Education. (2023). *Why does attendance matter for student success?*
<https://www.panoramaed.com/blog/why-does-attendance-matter-for-student-success>
- SchoolAnalytix. (2025). *Identifying at-risk students using data analytics*.
<https://www.schoolanalytix.com/identifying-at-risk-students-using-data-analytics-2/>
- SeatsOne. (2025). *The importance of early alerts in higher education*.
<https://seatsone.ai/the-importance-of-early-alerts-in-higher-ed/>
- Selvaraj, P., & Radhakrishnan, R. (2024). *Interactions of attendance and student performance: A quantitative analysis*. *DreamClass Research Insights*.
<https://www.dreamclass.io/2025/how-do-attendance-and-student-performance-affect-each-other/>
- U.S. Department of Education. (2023). *Early Warning Systems: Strategies and solutions*.
<https://www.attendanceworks.org/research/evidence-based-solutions/>

- Woods, K. (2024). *Using predictive analytics to identify at-risk students*. LinkedIn Learning Article. <https://www.linkedin.com/advice/0/what-most-effective-data-driven-strategies-identifying-ejj5f>
- World Bank. (2021). *Scalable early warning systems for school dropout prevention: Evidence from a randomized trial*. <https://documents1.worldbank.org/curated/en/983591622568486300/pdf/Scalable-Early-Warning-Systems-for-School-Dropout-Prevention-Evidence-from-a-4-000-School-Randomized-Controlled-Trial.pdf>
- Association for Supervision and Curriculum Development (ASCD). (2023). *Early warning systems for schools: Final report*. Education Resource Strategies. https://www.erstrategies.org/wp-content/uploads/2023/12/Early_Warning_Systems_for_Schools_FINAL_FINAL.pdf
- Bal, H., & Singh, G. (2023). *Improving real-time intervention through psychological early warning indicators: A systematic review*. *Frontiers in Psychology*, 14, 1189283. <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2023.1189283/full>
- Bono, S., & Sundt, J. (2025). *Impact of learning analytics–driven interventions on student persistence*. *SAGE Open*, 15(1), 1–14. <https://journals.sagepub.com/doi/full/10.1177/21582440251336707>
- Centre for Transforming Access and Student Outcomes (TASO). (2024). *Impact of student support interventions using learning analytics*. https://cdn.taso.org.uk/wp-content/uploads/2024_TASO_Impact-student-support-interventions-learning-analytics.pdf

- Chiu, T. K., & Hew, K. F. (2023). *Effects of timely feedback in learning analytics interventions: A meta-analysis*. *Frontiers in Psychology*, 13, 1096337. <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.1096337/full>
- Dahal, S., & Adhikari, K. (2024). *Predictive analytics and rapid intervention tools in higher education*. Semantic Scholar. <https://pdfs.semanticscholar.org/5cbf/2ce35e4bd5dd3fc6abe520b7b4b2ebca85e6.pdf>
- Digital Promise. (2024). *Timely learning analytics interventions: Case studies across institutions*. <https://scholars.hkmu.edu.hk/en/publications/learning-analytics-intervention-a-review-of-case-studies/>
- Education Resource Strategies. (2023). *Early warning systems for schools: Final report*. https://www.erstrategies.org/wp-content/uploads/2023/12/Early_Warning_Systems_for_Schools_FINAL_FINAL.pdf
- Federal Policy Institute. (2024). *Early warning systems for dropout prevention in the Philippines: Mobile-enabled solutions*. GSMA. https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/wp-content/uploads/2023/11/PhilippinesEWS_R_Web.pdf
- Institute of Education Sciences (IES). (2025). *Getting students on track for graduation: Impacts of the Early Warning Intervention and Monitoring System after one year*. <https://ies.ed.gov/use-work/resource-library/report/impact-study/getting-students-track-graduation-impacts-early-warning-intervention-and-monitoring-system-after-one>

- Institute for Labor Development and Analytics (ILDA). (2024). *Education dropout early warning case study*. https://ilda.la/wp-content/uploads/2025/04/2024-05-31_IDRC_ILDA-Case-Study-3_Education-Dropout-31-May-2024.pdf
- IARJSET. (2025). *Improving student monitoring through real-time analytics dashboards*. *International Advanced Research Journal in Science, Engineering and Technology*, 12(5). <https://iarjset.com/wp-content/uploads/2025/05/IARJSET.2025.125161.pdf>
- Learning Analytics Research Network (LARN). (2024). *Learning analytics interventions and their timing: Systematic review*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9901299/>
- Northern Illinois University. (2020). *Instructor perceptions of learning analytics alerts and their role in timely intervention* (Master's thesis). <https://huskiecommons.lib.niu.edu/cgi/viewcontent.cgi?article=8384&context=all-graduate-thesesdissertations>
- Penn State University. (2024). *Course Insights: Real-time dashboards now accessible to all instructors*. <https://www.psu.edu/news/faculty-and-staff/story/course-insights-now-available-all-instructors-penn-state>
- REL Northwest. (2025). *Supporting student success through early warning systems: A descriptive study*. <https://ies.ed.gov/rel-northwest/2025/01/descriptive-study-2>
- Russell, J., & Swanson, L. (2024). *Learning analytics impact on intervention timing: A meta-synthesis*. *SAGE Open*, 14(2). <https://journals.sagepub.com/doi/full/10.1177/21582440251336707>
- State of Colorado Department of Education. (2023). *Dropout prevention framework and early warning overview*. <https://www.cde.state.co.us/dropoutprevention/dpframeworkewsoverview>

University of Melbourne. (2022). *Learning analytics intervention timing and student persistence in higher education*. <https://iru.edu.au/wp-content/uploads/2022/05/MRUN-Learning-Analytics-report.pdf>

Williams, A., & Torres, L. (2025). *AI-enabled early alerts for preventive student support in higher education: A review*. *Education Data Science*, 6(4), 233–251. <https://edworkingpapers.com/sites/default/files/ai23-701.pdf>

Accessibility of student records

Formative/JMIR Publications. (2025). *Improving record accessibility through mobile-first learning analytics dashboards*. *JMIR Formative Research*, 2025(1), e70327. <https://formative.jmir.org/2025/1/e70327>

IARJSET. (2025). *Implementing real-time academic dashboards to improve data accessibility*. *International Advanced Research Journal in Science, Engineering and Technology*, 12(5). <https://iarjset.com/wp-content/uploads/2025/05/IARJSET.2025.125161.pdf>

Innovare SIP. (2025). *Common barriers to high-quality data in K–12 education and strategies to overcome them*. <https://innovaresip.com/blog/common-barriers-to-high-quality-data-in-k-12-education-and-strategies-to-overcome-them/>

Ifenatuora, A. C., Abdullahi, M. B., & Nwogu, E. C. (2023). *Accessibility evaluation of learning management systems in higher education*. *International Journal of Multidisciplinary Education*, 3(1), 23–35. https://www.allmultidisciplinaryjournal.com/uploads/archives/20250610125202_MGE-2025-3-231.1.pdf

Means, B., Peters, V., & Gallagher, L. (2007). *Use of education data at the local level: From accountability to instructional improvement*. U.S. Department of Education,

Office of Planning, Evaluation and Policy Development.
<https://files.eric.ed.gov/fulltext/ED501547.pdf>

Mutimukwe, C., Klein, J., & Wolf, K. (2021). *Bridging the accessibility gap in learning analytics dashboards*. In *Proceedings of the Learning Analytics Conference (LAK)*.
<https://oro.open.ac.uk/45313/1/LAK%20228%20final.pdf>

Panorama Education. (2023). *What is SIS integration and why it matters for student data access?* <https://www.panoramaed.com/blog/what-is-sis-integration>

PowerSchool. (2023). *Using analytics and MTSS visibility to improve access to student records*. <https://www.powerschool.com/blog/intervention-mtss-analytics-visibility/>

Research Publish. (2016). *Student information system: Design and significance*.
<https://www.researchpublish.com/upload/book/Student%20Information%20System-3278.pdf>

Sari, P., & Raharjo, B. (2025). *Enhancing academic monitoring using integrated analytic dashboards*. *IJSRET*, 11(3). <https://ceur-ws.org/Vol-2985/paper3.pdf>

TSAI (International Journal of Advanced Computer Science and Applications). (2021). *A framework for improving accessibility and usability in student information systems*. *IJACSA*, 12(2).
<https://thesai.org/Publications/ViewPaper?Volume=12&Issue=2&Code=IJACSA&SerialNo=10>

1st paragraph INTRO

IARJSET. (2025). *Analytics-driven academic monitoring system for identifying at-risk learners using dashboard visualization*. *International Advanced Research Journal in Science, Engineering and Technology*, 12(5). <https://iarjset.com/wp-content/uploads/2025/05/IARJSET.2025.125161.pdf>

- IrJIET. (2024). *Automated attendance and academic analytics system using integrated digital tools. International Research Journal of Innovations in Engineering and Technology*, 8(4).
https://irjiet.com/common_src/article_file/1725607181_a8f7245d03_8_irjiet.pdf
- Okokpujie, K., Noma-Osaghae, E., John, S., & Okokpujie, I. (2018). *A QR-code and RFID-based attendance monitoring system with authentication features for higher education institutions. Journal of Engineering Science and Technology*, 13(2), 350–366. <https://www.sciencedirect.com/science/article/pii/S2090447918300546>
- Raj, A., Gupta, P., & Mishra, R. (2024). *Automated attendance and analytics system using mobile QR code technologies. International Journal of Research in Applied Science & Engineering Technology*, 12(7). <https://www.ijraset.com/research-paper/automated-attendance-and-analytics-system>
- RSIS International. (2023). *QR code-based attendance management system. International Journal of Research and Innovation in Social Science*, 7(6).
<https://rsisinternational.org/journals/ijriss/articles/qr-code-based-attendance-management-system/>
- Sharma, S., Singh, R., & Patel, D. (2023). *Automated attendance systems for improved classroom management and data accuracy. IRJMETS*, 5(1).
https://www.irjmets.com/upload_newfiles/irjmets70500125547/paper_file/irjmets70500125547.pdf
- vMedulife. (2023). *Student attendance monitoring system: A smarter approach to classroom management. vMedulife Academic Blog*.
<https://vmedulife.com/blog/academic-planning/student-attendance-monitoring-system-a-smarter-approach-to-classroom-management/>

- De La Salle University. (2020). *Analytics-based student monitoring framework for academic risk identification*. Animo Repository. https://animorepository.dlsu.edu.ph/cgi/viewcontent.cgi?article=1004&context=conf_archers
- EJournals. (2019). *Automated attendance monitoring system: A Philippine school-based implementation study*. <https://ejournals.ph/article.php?id=15528>
- IIARI – International Institute of Academic Research and Industry. (2022). *RFID-based attendance monitoring system with SMS notification and data analytics*. In *Information Technology: Cross-Platform Application and Development*. https://iiari.org/online_book/information-technology-cross-platform-application-and-development/rfid-based-attendance-monitoring-system-with-sms-notification-and-data-analytics/
- Lyceum of the Philippines University–Laguna (LPU–Laguna). (2016). *LPU-L Student Electronic Attendance and Logging System (LPU-Laguna SEALS)*. <https://lpulaguna.edu.ph/wp-content/uploads/2016/08/Lpu-L-Student-Electronic-Attendance-And-Logging-System-Lpu-Laguna-Seals.pdf>
- WJARR – World Journal of Advanced Research and Reviews. (2023). *Web-based smart attendance monitoring system using digital technologies in Philippine schools*. <https://wjarr.com/sites/default/files/WJARR-2023-2644.pdf>